

Unit 1: Programming (scripting) Bootcamp

Table of contents

1	Getting Started	2
1.1	Command line interface (REPL) - using Julia as a calculator	2
1.2	Generating output - Hello World!	2
1.3	Using code editors	3
1.4	Input a variable through user interaction	3
2	Variables and types	3
2.1	Variables of type String (i.e. text) and Char(acter)	5
2.1.1	Operations with Variables	6
3	Built-in functions and macros	9
4	Logical statements (AND, OR, etc..)	11
4.1	Logical operators for composite types	13
5	Conditional execution	14
5.1	If statements (if-elseif-else)	14
5.1.1	Compact ways of formulating if clauses	15
6	Loops	15
6.1	For loops	17
6.2	While loops	19
6.3	break and continue	21
6.4	Howto implement do-while loops in Julia	22
7	Creating your own functions with (function)	23
7.1	Alternative, more compact ways, of defining functions	25
7.2	Arrays = Data	26
7.2.1	Indexing	26
7.3	Classroom Micro-Projects	29
7.4	Practice questions	30

In this unit we'll study the very basics of programming using the [Julia](#) language, though the concepts apply to pretty much every major programming language with slight modifications to the syntax. The unit ends with suggested class micro-projects and practice questions for the upcoming quiz. The pace and nature of this unit is aimed at students with little or no programming experience. The units afterwards pick up the pace significantly.

What to expect from this unit:

- Variables
- Types
- Using in-built functions (e.g. `println`) and macros (e.g. `@show`)
- Logical statements (AND, OR, etc..)

- Conditional statements (`if-elseif-else`)
- For loops
- While loops
- Creating your own functions with (`function`)

1 Getting Started

Install Julia for your particular type of computer following the steps listed on the [Julia website](https://docs.julialang.org). Then open a terminal on your computer and start Julia with the command ‘julia’. This should give you about the following output.

```
dietmar@Nina:~/github/MATH2504/ProgrammingCourse-with-Julia-SimulationAnalysisAndLearningSystems$ julia
```

```

      _       _       _
     (_      | (_   (_   | Documentation: https://docs.julialang.org
      _      | (_   (_   | Type "?" for help, "]"?" for Pkg help.
     _ _ _ _ | | _ _ _ _ |
    | | | | | | | / _ _ |
    | | | | | | | (_ _ |
   _/ | \ _ ' | _ _ | \ _ ' | Official https://julialang.org release
  |__/_/

```

```
julia>
```

1.1 Command line interface (REPL) - using Julia as a calculator

The prompt `julia>` represents the Julia command line interface called **REPL** (Read-Eval-Print-Loop). Start by **using it as a calculator**, e.g. at the Julia prompt enter

```
5+6
```

```
11
```

```
or
```

```
sin(25/180)
```

```
0.13844278873711527
```

Here the function `sin` from the standard library is called.

1.2 Generating output - Hello World!

Another function from the standard library can be called as follows:

```
println("Hello World!")
```

```
Hello World!
```

We are calling (or invoking) the function `println` with the argument `"Hello World!"`. The name of the function refers to ‘Print (and go into the next) line’. It takes the argument (the text `"Hello World!"` in this case) and writes it into the next line after the command, finally ending the line and moving the cursor into the next line.

Round brackets after a name are the application of a function, kind of like in mathematics. Keep in mind that round brackets also have other uses: For example, dealing with the order of precedence of computations:

```
1 + 1 * 4
```

```
5
```

as compared to

```
(1 + 1) * 4
```

8

or something more involved such as

```
(1/sin(25/180)+2.3)
```

9.523200349560057

1.3 Using code editors

The *REPL* is great for using Julia as a calculator. **Multi-line input (scripts, programs)** can be **pasted line-by-line and sometimes even at once directly into the REPL**, but it is more convenient to **store it into a .jl file**, and then

- invoke the file line by line either from the terminal running ‘julia filename.jl’,
- or from the REPL using ‘include(“filename.jl”)’.

Or, you use a **dedicated editor**, e.g.

- install [VSCode](#) and, once you started the editor, install the Julia language extension. (If you can’t make it work, we’ll show you in the tutorials how to do this) and
- the [Jupyter notebook interface](#) which can be installed by running, line-after-line, in the REPL

```
using Pkg
Pkg.add("IJulia")
using IJulia
notebook()
```

1.4 Input a variable through user interaction

Store the following 3 lines into the file ‘test.jl’ and run it either by running the command ‘julia test.jl’ from the terminal or through include(“test.jl”) from the REPL, or directly from VSCode.

```
println("What is your name: ")
username=readline()
println("Hi ", username, ", how are you?")
```

This short script outputs the question about your name, and then used the function **readline** to read in your name and store it into the variable **name**. Finally it outputs a greeting which includes your particular name.

Try running this simple script in several ways:

- REPL (by running Julia and using the REPL, line after line)
- Jupyter (in a Jupyter notebook with Ctrl+Enter),
- In a file (by creating a file such as my_code.jl and either running `julia my_code.jl` from the terminal or include(“my_code.jl”) within the REPL)
- In VS-Code with the Julia extension.

2 Variables and types

In the last example the variable **username** acted as a container for the name the user would input. Indeed, one may think of variables as a **storage box**, with a name (tag) on it. The computer’s memory is the **storage facility** in which you can store all your boxes.



Figure 1: Variables are boxes in a storage facility (memory)

For example the command

```
x = 3
```

3

stores the value 3 in the box tagged **x**.

Later, you may retrieve the value stored in the variable **x**, for example the command

```
x + 25
```

28

retrieves the value stored in **x** and adds 25 to it.

Variables have a type - think of it as the size and shape of the box. The command

```
typeof(x)
```

Int64

gives the type of the variable **x** (an integer value of size 8 bytes). The size can be obtained using the function **sizeof**, e.g.

```
sizeof(x)
```

8

The type Int64 is the standard type for integers. Particular integer numbers are written without **.**:

```
typeof(32)
```

Int64

Another big class of numeric variables are floating point numbers. Specific floating point numbers are written with at **.**, e.g.

```
y=34.23
typeof(y)
```

Float64

```
sizeof(y)
```

8

defines a Floating point number of size 8 bytes. Floating point numbers are also the result when we divide integer numbers, e.g.

```
r=3/4
println(r)
typeof(r)
```

0.75

Float64

and the result of many functions which typically return floating point numbers such as

```
result=cos(2)
println(result)
typeof(result)
```

-0.4161468365471424

Float64

2.1 Variables of type String (i.e. text) and Char(acter)

```
str = "Hello"
```

"Hello"

```
typeof(str)
```

String

```
sizeof(str)
```

5

```
isbits(str)
```

false

```
isbits(x)
```

true

x is a **primitive type**, i.e. x **directly** refers to the value of the variable.

String is the standard data type for text. It is a **composite type**. Therefore the variable 'str' does not refer to the text "Hello" itself, but to the address (**reference**, pointer) in memory where the content is stored. Think of the reference as the shelf number in the storage facility (memory).

```
y = " " # \eta + [TAB]
```

" "

```
sizeof(y)
```

2

```
y = "p"
```

"p"

```
sizeof(y)
```

1

```
y = 'p'
```

'p': ASCII/Unicode U+0070 (category Ll: Letter, lowercase)

```
typeof(y)
```

Char

```
isbits(y)
```

true

2.1.1 Operations with Variables

```
x = 3.0
```

```
y = 4.2
```

```
x*y, x+y, x-y, x/y, x^y #In Python it would have x**y (raising to a power)
```

(12.600000000000001, 7.2, -1.2000000000000002, 0.7142857142857143, 100.90420610885693)

And for non-numerical types such as Strings:

```
s1 = "hello " #notice the extra space
```

```
s2 = "world"
```

```
@show typeof(s1);
```

```
s1*s2 #In Python it would have been x+y for concatenation
```

```
typeof(s1) = String
```

"hello world"

```
s = "hello "
```

```
y = 5
```

```
x^y
```

243.0

These are examples of **operator (method) overloading** in Julia: For different data types, different versions of a function/method/operator can be defined.

2.1.1.1 Indexing into strings using [...]

```
s[1:end-1]
```

"hello"

```
s^(y-1)*s[1:end-1]
```

"hello hello hello hello hello"

Since the operator '^' has precedence of '*', this is the same as

```
(s^(y-1))*s[1:end-1]
```

"hello hello hello hello hello"

2.1.1.2 Common errors: type mismatch

What if we did the brackets the other way?

```
s^((y-1)*s[1:end-1])
```

```
MethodError: MethodError(*, (4, "hello"), 0x000000000000097ae)
```

```
MethodError: no method matching *(::Int64, ::String)
```

The function `\*` exists, but no method is defined for this combination of argument types.

Closest candidates are:

```
*(::Any, ::Any, ::Any, ::Any...)
```

```
@ Base operators.jl:642
```

```
*(::Missing, ::Union{AbstractChar, AbstractString})
```

```
@ Base missing.jl:174
```

```
*(::Real, ::Dates.Period)
```

```
@ Dates
```

```
~/.julia/juliaup/julia-1.12.6+0.x64.linux.gnu/share/julia/stdlib/v1.12/Dates/src/periods.jl:91
```

```
...
```

Stacktrace:

```
[1] top-level scope
```

```
@
```

```
~/github/ProgrammingCourse-with-Julia-SimulationAnalysisAndLearningSystems/quarto/lecture-unit-1.qmd:280
```

Here, Julia tries to identify an instance of the method “+(:,)” which takes an Integer and a String as arguments (this is Julia’s “**multiple dispatch**” paradigm), but doesn’t succeed. The closest candidates are listed in the error message, but none of them fits (e.g. any argument of type Int64 can be converted into types Real, Number, Any, etc).

```
x = 2
```

```
√x #\sqrt + [TAB]
```

```
1.4142135623730951
```

```
sqrt(x)
```

```
1.4142135623730951
```

```
round(√x, digits = 3)
```

```
1.414
```

```
y = -x
```

```
-2
```

```
sqrt(y)
```

```
DomainError: DomainError(-2.0, "sqrt was called with a negative real argument but will only return a complex result if called with a complex argument. Try sqrt(Complex(x)).")
```

```
DomainError with -2.0:
```

sqrt was called with a negative real argument but will only return a complex result if called with a complex argument. Try sqrt(Complex(x)).

Stacktrace:

```
[1] throw_complex_domainerror(f::Symbol, x::Float64)
```

```
@ Base.Math ./math.jl:33
```

```
[2] sqrt
```

```
@ ./math.jl:627 [inlined]
```

```
[3] sqrt(x::Int64)
```

```
@ Base.Math ./math.jl:1546
```

```
[4] top-level scope
```

```
@
```

```
~/github/ProgrammingCourse-with-Julia-SimulationAnalysisAndLearningSystems/quarto/lecture-unit-1.qmd:303
```

```
z = sqrt(y + 0im)
```

```
0.0 + 1.4142135623730951im
```

2.1.1.3 Be careful about side effects when manipulating composite types

For primitive types we have

```
a1=1; a2=a1; a1=99;  
println("After changing the value of a1, the value of a2 is: ", a2)
```

After changing the value of a1, the value of a2 is: 1

but for a composite type such as `Vector{Float64}`

```
v1=[1,2,3]; v2=v1; v1[1]=99;  
println("After changing the value of v1, the value of v2 is: $v2")
```

After changing the value of v1, the value of v2 is: [99, 2, 3]

2.1.1.4 Syntactic sugar for swapping values

```
x = 20  
y = 5  
  
#swapping x and y: attempt 1  
x = y  
y = x  
  
@show x; #This is the @show macro  
@show y;
```

```
x = 5  
y = 5
```

```
x = 20  
y = 5  
  
#swapping x and y: attempt 2  
temp = x  
x = y  
y = temp  
  
@show x;  
@show y;
```

```
x = 5  
y = 20
```

In Julia we can do it without using the variable `temp`!

```
x = 20  
y = 5  
  
x, y = y, x #Julia specific solution  
  
@show x;  
@show y;
```

```
x = 5
```


y = 20

Now, with composite types:

```
m1 = rand(1000, 1000) #generates of 1000x1000 matrix with random entries
m2 = rand(1000, 1000)

@time m2, m1 = m1, m2;
```

0.000013 seconds (46 allocations: 2.141 KiB)

takes about as much time as

```
m1 = rand(10000, 10000)
m2 = rand(10000, 10000)

@time m2, m1 = m1, m2;
```

0.000014 seconds (9 allocations: 464 bytes)

because for composite types **only the references** are swapped! Compare this to the time required for re-assigning those variables after copying their entire content:

```
m1 = rand(1000, 1000)
m2 = rand(1000, 1000)

@time m2, m1 = copy(m1), copy(m2);
```

0.005428 seconds (19 allocations: 15.260 MiB)

3 Built-in functions and macros

So we have seen in-built functions like `println` and `sqrt`, but surely there are many more?

```
println("hello world")
```

hello world

The `println` function can also take several arguments.

```
println("hello", " ", "world")
```

hello world

Here is `print` without a new line.

```
print("hello")
print(" ")
print("world")
```

hello world

```
println("hello\nworld") #notice the \n (newline)
println("Here is the next line")
```

hello

world

Here is the next line

An equivalent use of `print`:

```
print("hello")
print("\n")
print("world")
print("\n")
println("Here is the next line")
```

```
hello
world
Here is the next line
```

Let's see some mathematical in-built functions:

```
exp(1)
```

```
2.718281828459045
```

```
#\euler + [TAB]
```

```
= 2.7182818284590...
```

```
log(^2), log2(2^5), log10(10^3)
```

```
(2.0, 5.0, 3.0)
```

```
abs(-3.5), abs(3.5), abs(3 + 4im)
```

```
(3.5, 3.5, 5.0)
```

But you can't do absolute value of a string:

```
abs("3.5")
```

```
MethodError: MethodError(abs, ("3.5",), 0x000000000000097b6)
```

```
MethodError: no method matching abs(::String)
```

```
The function `abs` exists, but no method is defined for this combination of argument types.
```

```
Closest candidates are:
```

```
abs(::Bool)
```

```
@ Base bool.jl:155
```

```
abs(::Pkg.Resolve.VersionWeight)
```

```
@ Pkg
```

```
~/julia/juliaup/julia-1.12.6+0.x64.linux.gnu/share/julia/stdlib/v1.12/Pkg/src/Resolve/versionweights.jl:32
```

```
abs(::Missing)
```

```
@ Base missing.jl:101
```

```
...
```

```
Stacktrace:
```

```
[1] top-level scope
```

```
@
```

```
~/github/ProgrammingCourse-with-Julia-SimulationAnalysisAndLearningSystems/quarto/lecture-unit-1.qmd:445
```

```
length("3.5")
```

```
3
```

```
factorial(4)
```

```
24
```

Note that '@show' does something very similar. It is a **macro** (all macros have @ in their names).

```
s="3.5";
@show s;
println(s)
```

```
s = "3.5"
3.5
```

Macros are evaluated at compile time (when translating the Julia code into machine code) and therefore have access to more information, e.g. the variable name of their argument. For that reason `@show` can also output the variable name ‘s’ which is convenient for debugging, whereas `println` only has access to the content ‘3.5’ of its argument. `##` Type conversions

```
a=4;
x=Float64(a);
@show typeof(a) a;
@show typeof(x) x;
```

```
typeof(a) = Int64
a = 4
typeof(x) = Float64
x = 4.0
```

```
y=sqrt(5);
b=round(y);
@show typeof(y) y;
@show typeof(b) b;
```

```
typeof(y) = Float64
y = 2.23606797749979
typeof(b) = Float64
b = 2.0
```

```
Int('a'), Int('b'), Int('A'), Int('B'), Int(' '), Int(' ') #type \alpha and \forall
```

```
(97, 98, 65, 66, 945, 8704)
```

gives the ascii/unicode for these letters/symbols.

Often those conversions are made implicitly (“**multiple dispatch**”), e.g.

```
z=a+x
@show typeof(z);
```

```
typeof(z) = Float64
```

4 Logical statements (AND, OR, etc..)

```
true, false #In Python it is True and False (caps first letter)
```

```
(true, false)
```

```
typeof(true)
```

```
Bool
```

```
@show Int(true);
@show Int(false);
```

```
Int(true) = 1
Int(false) = 0
```

4.0.0.1 Logical operators

```
5 == 2 + 3 # check for equality
```

```
true
```

```
5 != 2 + 3 # check for not being equal (!=)
```

```
false
```

```
false && false, false && true, true && false, true && true #logical AND
```

```
(false, false, false, true)
```

```
false || false, false || true, true || false, true || true #logical OR
```

```
(false, true, true, true)
```

```
(2 != 3) || (2 == 3)
```

```
true
```

```
!(2 == 3) # ! (not)
```

```
true
```

For the next block, you need to have the package `Random` installed: To install it, change to package mode using `]`, and use `add Random`.

The REPL modes are `*` ; to access the shell/terminal temporarily , `* ?` for help mode `* ]` for package mode `* Ctrl-C` or `Backspace` brings you back to the standard “Julian” prompt.

```
using Random
```

```
Random.seed!(7) #The exclamation mark denotes functions which change the environment  
(side-effects).
```

```
x = rand(1:100) #random number within 1, 2, ..., 100
```

```
y = rand(1:100) #random number within 1, 2, ..., 100
```

```
@show x, y;
```

```
(x == y) || (x != y)
```

```
(x, y) = (25, 72)
```

```
true
```

The command `Random.seed!` sets the seed value for the random number generator. Without it, an almost random seed value is generated and random numbers will differ every time you evaluate the following ‘rand’ commands. If you set the seed value, then random numbers will always be the same - every time you run the script (which can be useful for debugging, or comparison).

```
x < y, x<=y, x > y, x >= y # \le or \ge + [TAB]
```

```
(true, true, true, false, false)
```

```
(x == y) && (x != y)
```

```
false
```

4.1 Logical operators for composite types

```
x=[1,2,3]  
y=[1,2,3]  
x==y
```

true

compares the content of x to the content of y. To check whether x and y are physically identical (same reference) use

```
@show x===y;  
z=y  
@show z===y;
```

```
x === y = false  
z === y = true
```

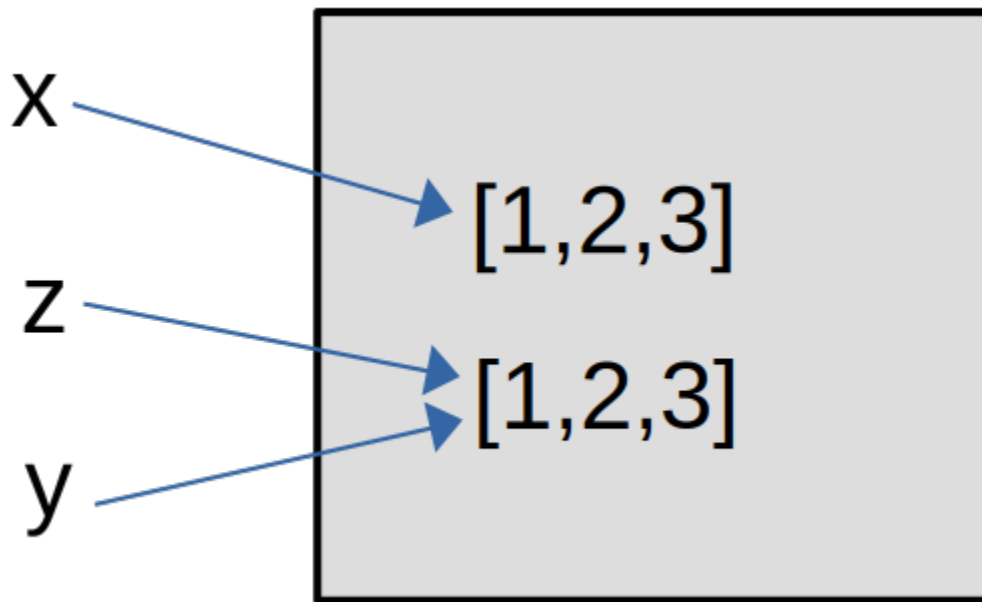
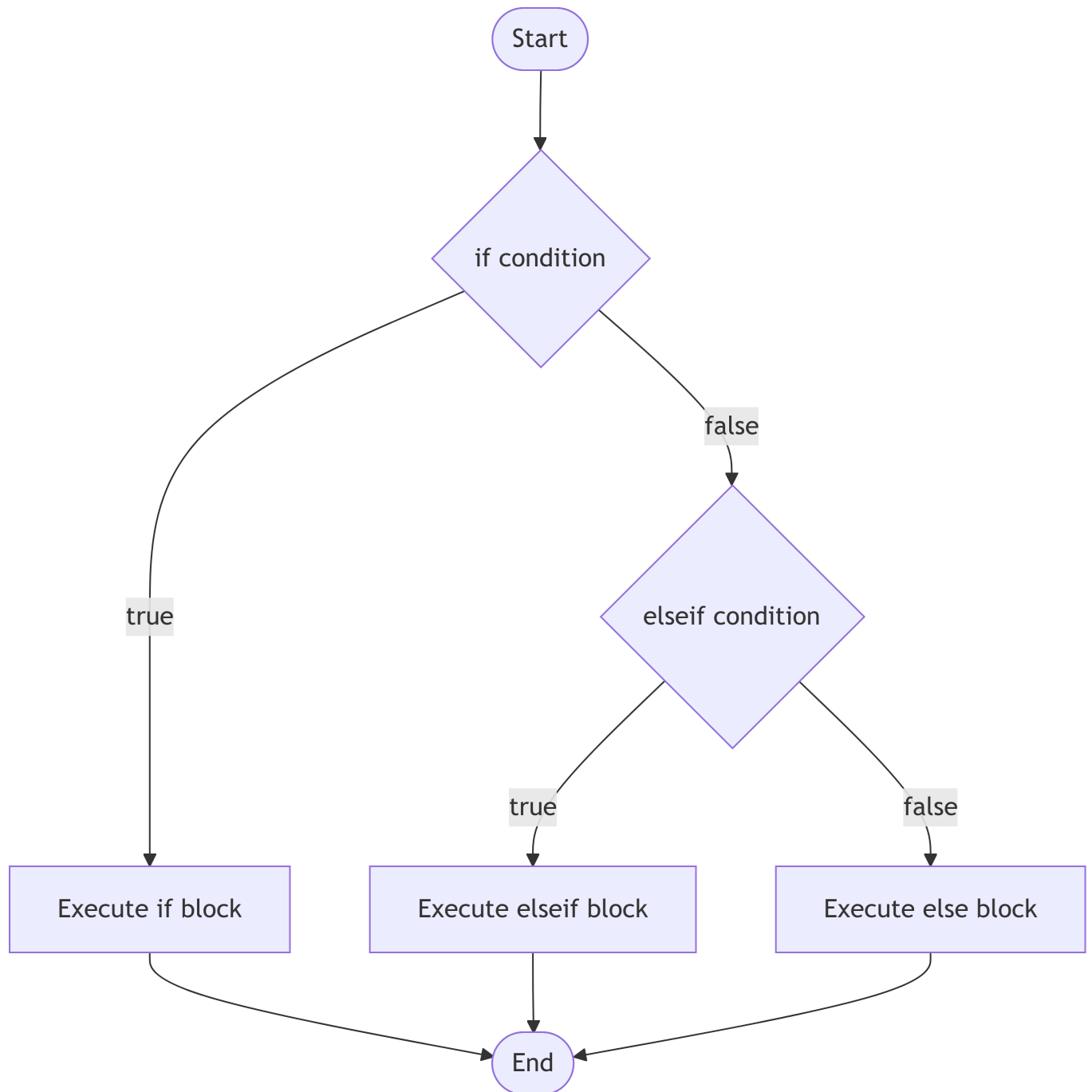


Figure 2: composite types

5 Conditional execution

5.1 If statements (if-elseif-else)



```
if 2 < 3 && 2 > 3
  println("The world has gone crazy")
end
```

```
if 2 < 3 || 2 > 3
  println("The world makes sense")
end
```

```
end
```

The world makes sense

```
x = 25.3
if x < 30
    println("It is less than 30")
else
    println("It is greater or equal to 30")
end
```

It is less than 30

```
x = 25.3
if x < 20
    println("It is less than 20")
elseif x < 30
    println("It is less than 30 but not less than 20")
else
    println("It is greater or equal to 30")
end
```

It is less than 30 but not less than 20

Let's use it to compute an absolute value.

```
x = -3 # some input
if x < 0
    println(-x)
else
    println(x)
end
```

3

5.1.1 Compact ways of formulating if clauses

Use either the **ternary operator** `condition ? dothisiftrue : dothisiffalse`

```
x = -3 # some input
absx = x < 0 ? -x : x
```

3

or using logical AND (`&&`), making use of Julia's **lazy evaluation of logical operators**,

```
x = -3 # some input
x < 0 && println("This is a negative number!")
```

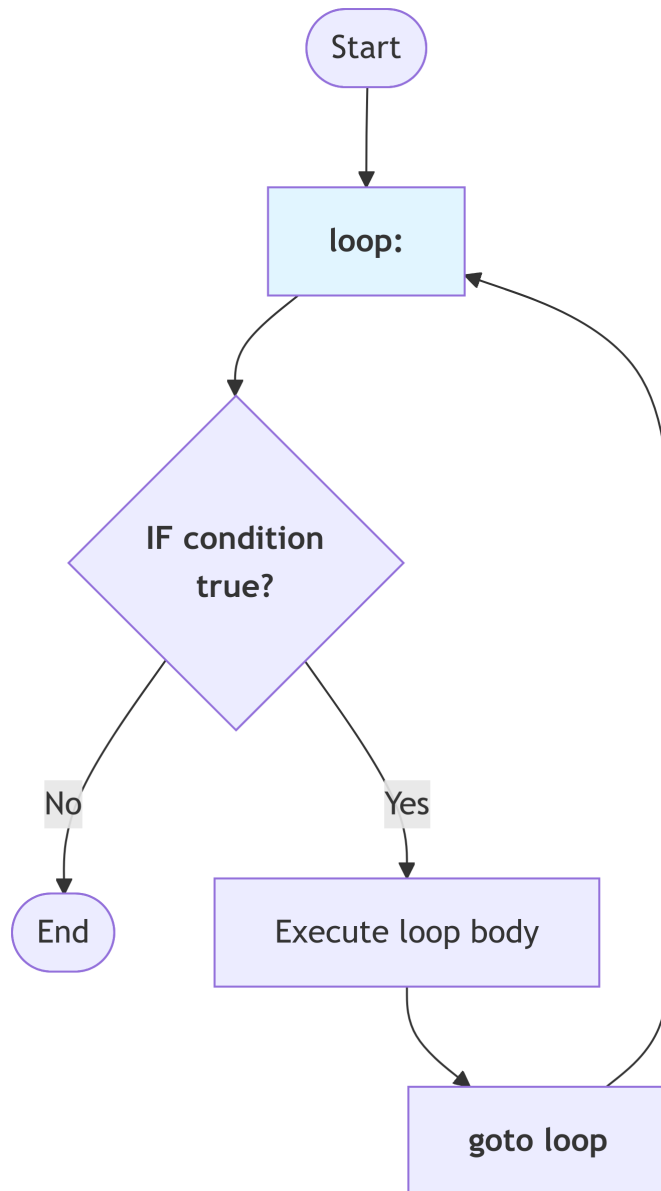
This is a negative number!

Here, the second condition for the AND clause `&&` is only evaluated if the first condition evaluates to true. Effectively this behaves like a single-line if statement.

6 Loops

You might want to execute a specific section of your code more than once, for example for a certain number of times, or until a condition is reached. The section of a program which is structured in this way is called a **loop**.

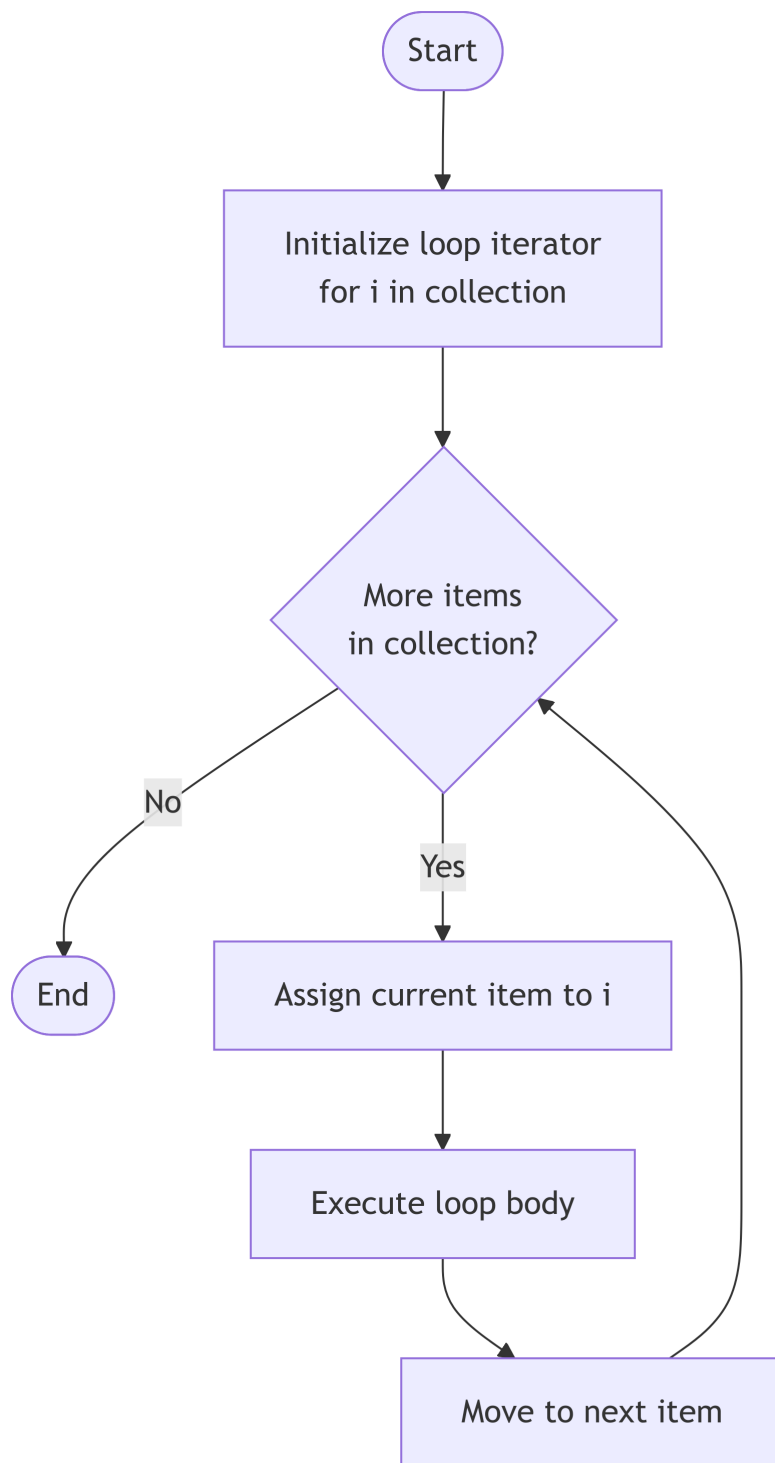
In principle, a loop can be implemented by combining an if-statement which checks whether the loop should be executed one more time or not, and a goto statement by which program execution jumps back to the beginning of the loop section.



In many programming languages, particularly older ones, such `goto` statements exist, but they make code error-prone as well as hard to read and to maintain. For that reason more modern programming languages either don't include any type of `goto` command, or just in the form of macros which is the case for Julia.

It is pretty much always better to implement loops through dedicated control structures, namely `for` loops and `while` loops.

6.1 For loops



```
for i in 1:5
    println(i)
end
```

```
2
3
4
5
```

```
for i in 1:3
    println(i)
end
```

```
1
2
3
```

```
for i 1:3 # \in + [TAB]
    println(i)
end
```

```
1
2
3
```

```
for _ 1:3 # \in + [TAB]
    println("hello")
end
```

```
hello
hello
hello
```

```
total = 0
max_val = 10
for i 1:max_val
    global total # This line can be ignored by now
    total = total + i
end
println("The total is: $total")
println("And using a formula: ", max_val*(max_val+1)/2)
```

```
The total is: 55
And using a formula: 55.0
```

Notice the above 55 vs. 55.0. This is because of division which converts an integer to a float.

Using the command **break**, loops can be left at any time.

```
for i 1:13 # \in + [TAB]
    if i<5
        println("hello $i")
    else
        break
    end
end
```

```
hello 1
hello 2
hello 3
hello 4
```

Using the command **continue**, code execution continues at the start of the loop

```

for i 1:13 # \in + [TAB]
  if i<10
    continue
  else
    println("hello $i")
  end
end
end

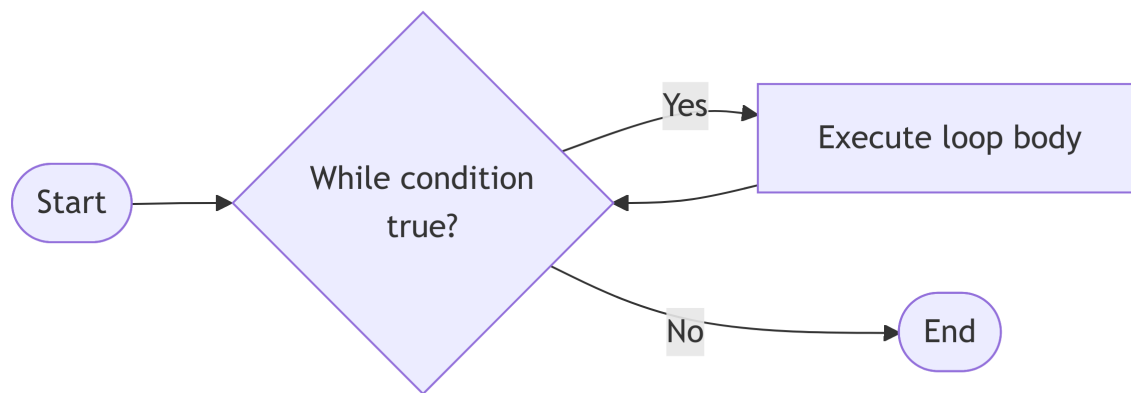
```

```

hello 10
hello 11
hello 12
hello 13

```

6.2 While loops



While loops give you more control over the loop.

```

i = 1
while i 3 # do the following as long as i is smaller or equal than 3
  global i
  println(i)
  i = i + 1
end

```

```

1
2
3

```

The hailstone sequence: If a number is even, half it, if it is odd, multiply by 3 and add 1. Stop when you reach 1.

```

n = 7
while n != 1
  global n
  print(n, ", ")
  if n % 2 == 0 # modulo
    n = n ÷ 2 # \div + [TAB] Integer division
  else
    n = 3n + 1
  end
end
println(n)

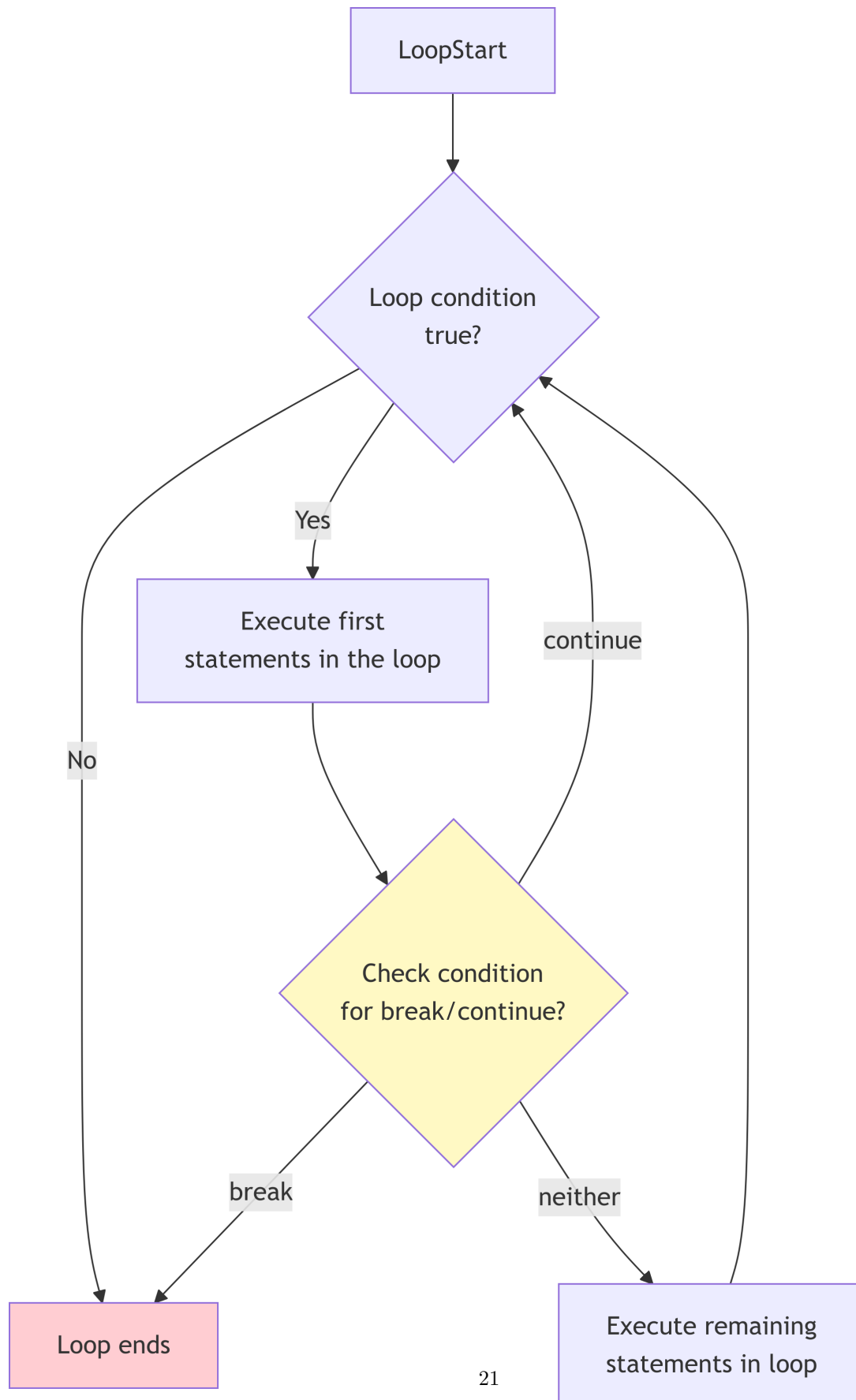
```

```

7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1

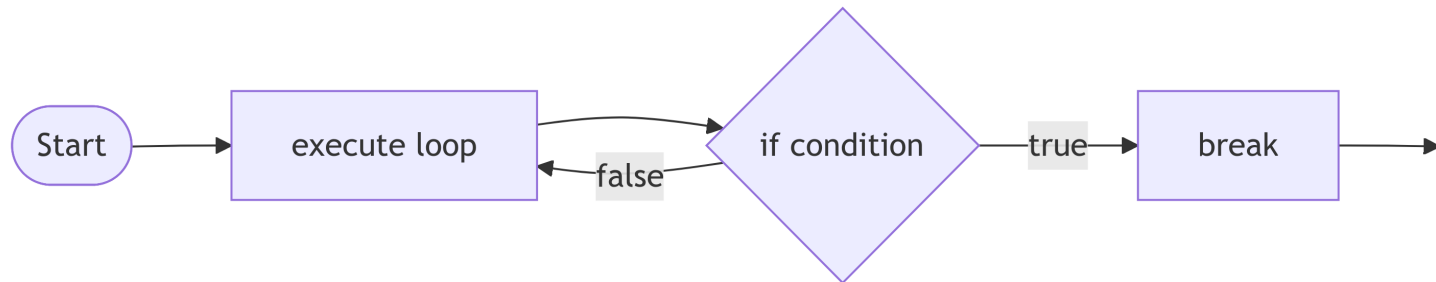
```


6.3 break and continue



Using the command `break`, loops can be left at any time.

6.4 Howto implement do-while loops in Julia



In do-while loops the loop is executed as least once before the termination condition is evaluated!

```
while true
    # do stuff
    if cond()
        break
    end
end
```

```
UndefVarError: UndefVarError(:cond, 0x000000000000097be, Main.Notebook)
```

```
UndefVarError: `cond` not defined in `Main.Notebook`
```

```
Suggestion: check for spelling errors or missing imports.
```

```
Hint: a global variable of this name also exists in LinearAlgebra.
```

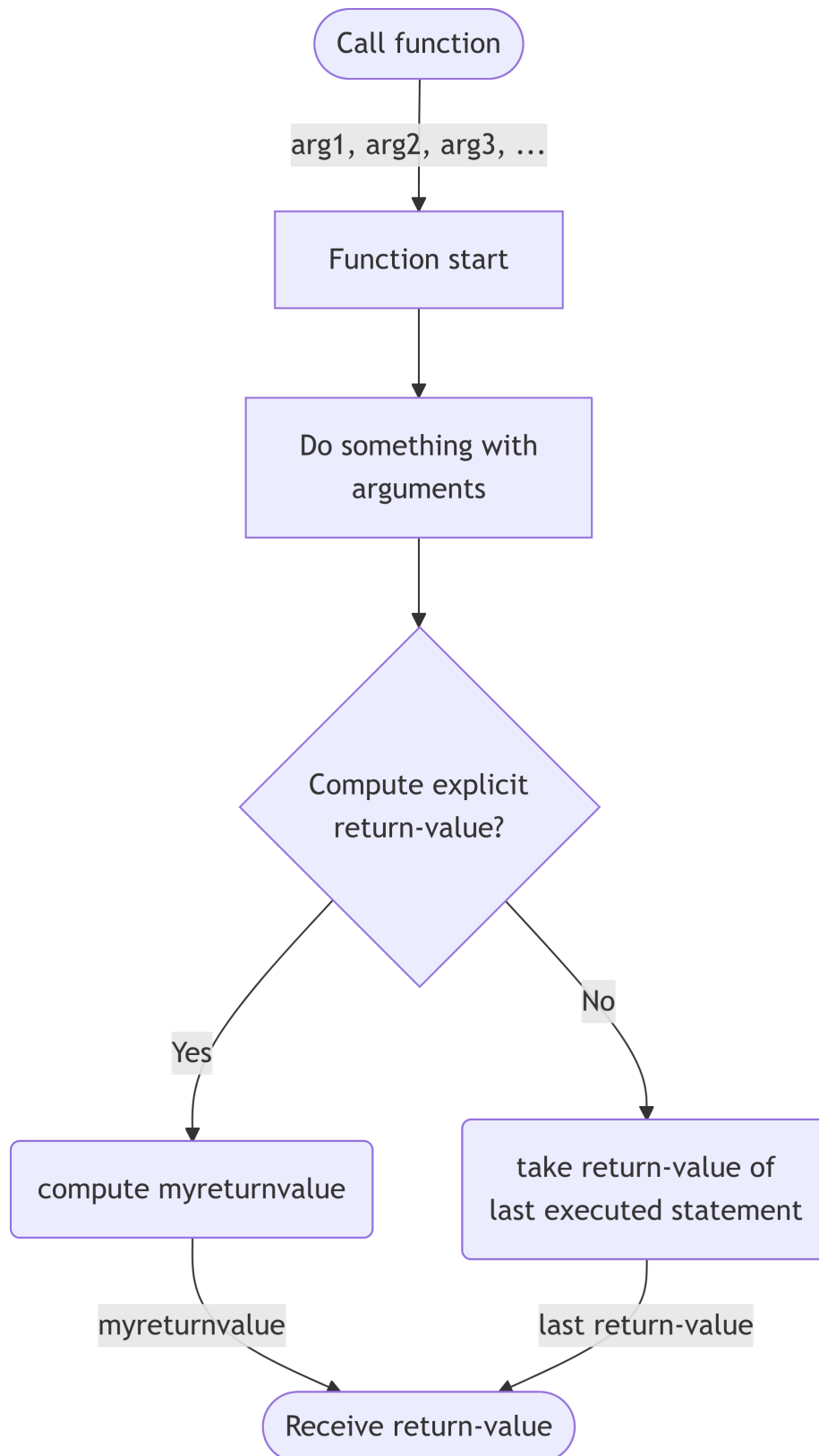
```
Stacktrace:
```

```
[1] top-level scope
```

```
@
```

```
~/github/ProgrammingCourse-with-Julia-SimulationAnalysisAndLearningSystems/quarto/lecture-unit-1.qmd:824
```

7 Creating your own functions with (function)



We already had logic for an absolute value function (of real values). Now let's make a function out of it:

```
function my_abs(x)
  if x < 0
    return -x
  else
    return x
  end
end
```

my_abs (generic function with 1 method)

We can also implement this as,

```
function my_abs(x)
  if x < 0
    return -x
  end
  return x
end
```

my_abs (generic function with 1 method)

Or even,

```
function my_abs(x)
  if x < 0
    return -x
  end
  x
end
```

my_abs (generic function with 1 method)

which uses that functions by default return the output of the last command.

```
@show my_abs(-3.5);
@show my_abs(2.3);
@show my_abs(0);
```

```
my_abs(-3.5) = 3.5
my_abs(2.3) = 2.3
my_abs(0) = 0
```

Functions don't have to have an argument or a return value:

```
function print_my_details()
  println("Name: Jacob")
  println("Occupation: diesel mechanic")
  return nothing
end

print_my_details()
```

Name: Jacob

Occupation: diesel mechanic

Let's see the hailstone sequence for the first 10 integers:

```
#let's "wrap" the code we had before in a function
function hailstone(n_start)
  n = n_start
  while n != 1
```



```

print(n, ", ")
if n % 2 == 0
  n = n ÷ 2
else
  n = 3n + 1
end
end
println(n)
return nothing
end

hailstone(7)

```

7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1

```

println("Hailstone sequences: ")
for n in 1:10
  hailstone(n)
end

```

Hailstone sequences:

```

1
2, 1
3, 10, 5, 16, 8, 4, 2, 1
4, 2, 1
5, 16, 8, 4, 2, 1
6, 3, 10, 5, 16, 8, 4, 2, 1
7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1
8, 4, 2, 1
9, 28, 14, 7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1
10, 5, 16, 8, 4, 2, 1

```

7.1 Alternative, more compact ways, of defining functions

When you can implement a function in one line, you can avoid using the `function` keyword and instead use **mathematical notation for defining functions**

```

times_2(x) = 2x

@show times_2(times_2(10)) #this uses the function twice.
@show (times_2 times_2)(10) # alternative way of writing function composition
@show [1,2,3] |> times_2; # or, using piping operator |>

```

```

times_2(times_2(10)) = 40
(times_2 times_2)(10) = 40
[1, 2, 3] |> times_2 = [2, 4, 6]

```

One can even define **anonymous function** without function names

```

[1,2,3] |> (x->2x)

```

```

3-element Vector{Int64}:
 2
 4
 6

```

7.2 Arrays = Data

So up to this point we have the basics of programming including variables, logical statements, conditional statements, loops, and functions. One other major component is **data**. For this let us consider arrays.

```
[2, 4, -3, 15] #this is an array
```

```
4-element Vector{Int64}:
 2
 4
-3
15
```

Note that in Julia 1-dimensional arrays are also called Vector. Arrays represent a “block of memory” (RAM). Its address in RAM (random access memory) can be retrieved as follows:

```
a = [2, 4, -3, 15]
b = a #This just makes b point to the same chunk of memory as a.
@show pointer(a);
@show pointer(b);
```

```
pointer(a) = Ptr{Int64}(0x00007f8e2c90b6a0)
pointer(b) = Ptr{Int64}(0x00007f8e2c90b6a0)
```

You can see that both a and b “point” to the same location in memory (a reference, or pointer).

7.2.1 Indexing

We can get an element of an array by “indexing”. In Julia (like MATLAB and R) indexing starts at 1. In languages like Python, Javascript, and C/C++, indexing starts at 0.

```
@show a[1];
@show a[2];
@show a[3];
@show a[4];
```

```
a[1] = 2
a[2] = 4
a[3] = -3
a[4] = 15
```

You can also change the elements this way.

```
a[3] = -100.
@show a;
```

```
a = [2, 4, -100, 15]
```

```
@show b;
```

```
b = [2, 4, -100, 15]
```

```
length(a)
```

```
4
```

Note that strings are kind of like arrays.

```
my_string = "Hello"
@show length(my_string)
@show my_string[1]
@show my_string[5];
```

```
length(my_string) = 5
my_string[1] = 'H'
my_string[5] = 'o'
```

But arrays are **mutable** (meaning you can change them) while strings are not (they are **immutable**).

```
my_string[5] = 'x'
```

```
MethodError: MethodError(setindex!, ("Hello", 'x', 5), 0x000000000000097cc)
MethodError: no method matching setindex!{::String, ::Char, ::Int64}
The function `setindex!` exists, but no method is defined for this combination of argument types.
Stacktrace:
 [1] top-level scope
      @
~/github/ProgrammingCourse-with-Julia-SimulationAnalysisAndLearningSystems/quarto/lecture-unit-1.qmd:1002
```

The error above is how Julia tries to tell you that you can't change the string.

Arrays don't have to be arrays of just numbers, for example.

```
my_other_array = ["hello", 2, sqrt, 3.4, "bye", [1, 2, 3]]
```

```
6-element Vector{Any}:
 "hello"
 2
 sqrt (generic function with 19 methods)
 3.4
 "bye"
 [1, 2, 3]
```

is an array of components of type `Any` (the most general type Julia provides).

Compare this to the definition of a tuple

```
my_other_array = ("hello", 2, sqrt, 3.4, "bye", [1, 2, 3]);
typeof(my_other_array)
```

```
Tuple{String, Int64, typeof(sqrt), Float64, String, Vector{Int64}}
```

Let's make a function that sums up entries of an array (assuming it is an array of numbers).

```
function my_sum(input_array)
    total = 0
    for i in 1:length(input_array)
        total += input_array[i]
    end
    return total
end

a = [1, 10, 100, 1000]
my_sum(a)
```

```
1111
```

There is also an in-built function for this.

```
sum(a)
```

```
1111
```

But you can do more things with the in-built sum. For example:

```
a = [-1,10,-100,1000]
@show sum(a)
@show sum(abs, a); #sum the absolute values
```

```
sum(a) = 909
sum(abs, a) = 1111
```

Arrays can be extended with the `push!` function. Note that the `!` is part of the function name and indicates to us that the function modifies the array.

```
my_array = Int[] #empty array of integers
push!(my_array, -3)
@show my_array
push!(my_array, 10)
@show my_array;
```

```
my_array = [-3]
my_array = [-3, 10]
```

Let's modify our `hailstone` function to return an array with the sequence instead of printing the sequence.

```
function hailstone(n_start)
    sequence = Int[]
    n = n_start
    while n != 1
        push!(sequence, n) # was before print(n, ", ")
        if n % 2 == 0
            n = n ÷ 2
        else
            n = 3n + 1
        end
    end
    push!(sequence, n) # was before println(n)
    return sequence # was before return nothing
end

hailstone(7)
```

```
17-element Vector{Int64}:
 7
22
11
34
17
52
26
13
40
20
10
 5
16
 8
 4
 2
 1
```

The [Collatz Conjecture](#) says that for every starting value the sequence eventually hits 1. Let's try to disprove

it by seeing if this program gets stuck.

```
for n in 1:10^3
    hailstone(n)
end
```

It didn't get stuck (try even changing 10^3 to 10^6).

Now let's see what was the longest sequence.

```
length_of_longest = 0
n_of_longest = 1
for n in 1:10^3
    global length_of_longest, n_of_longest
    seq_len = length(hailstone(n))
    if seq_len > length_of_longest
        length_of_longest = seq_len
        n_of_longest = n
    end
end

println("Longest sequence is of length $length_of_longest when you start at $n_of_longest.")
```

Longest sequence is of length 179 when you start at 871.

One very easy way to create an array is via an array comprehension.

```
[i^2 for i in 1:10]
```

10-element Vector{Int64}:

```
1
4
9
16
25
36
49
64
81
100
```

The `maximum` function finds the maximal value of an array. We can find the 179 value from above just like this.

```
maximum([length(hailstone(n)) for n in 1:10^3])
```

179

What if we wanted to also know which index attains this maximum? See the `findmax` function.

```
findmax([length(hailstone(n)) for n in 1:10^3])
```

(179, 871)

7.3 Classroom Micro-Projects

Here are several micro-projects aimed at providing practice using the concepts above. For each of them you may need a bit more functionality and in-built functions that what we explored so far. You can get that from web-search, LLM-help, peer help, or staff help.

Micro-Project 1: Write a little program in that gives a student a mathematics arithmetic quiz with the operations $+$, $-$ and $*$. It has input numbers in the range from 1 to 200. Students get asked questions and need to respond with the correct answer. After 10 questions the students gets a summary of how many they got correct and what they got wrong. You may need to explore the `readline` and `parse` functions.

Micro-Project 2: Write a little program that is a Tik-Tak-Toe game between two players. On each turn, the state of the board is re-printed. You can use 1,...,9 as input for the cells.

Micro-Project 3: Write a program that works on some fixed long string (e.g. several paragraphs of text such as the text here describing the micro projects). The program should output the character count, the word count, and perhaps other statistics about the text.

7.4 Practice questions

1. Consider the Julia code:

```
a = ["$i" for i in 1:3]
s = a[1]*a[2]*a[3]
```

What is the type and value of `s`?

2. You want to write a function `my_minimum` that gets an array of numbers and returns the minimal value in the array. Do not use the `minimum` in-built function as part of your answer.

```
function my_minimum(a)

    return minimal_value
end
```

Fill in the code for the function.

3. You want to write a Julia function `num_sub_str` which accepts a string `main_string` and a string `sub_string`, and returns a count of how many times `sub_string` is in `main_string`.

```
function num_sub_str(main_string, sub_string)

    return number_occurrences
end
```

Fill in the code for the function.

4. Consider the Julia function, `tamid_nahon` which gets two boolean values as inputs, `a`, and `b`.

```
function tamid_nahon(a, b)
    return !(a && b) == !a || !b
end
```

For what combinations of `a` and `b` is the return value `true`? For what is it `false`?

5. The Fibonacci sequence is 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, ... (every element is the sum of the previous two and the initialization is 0, 1). Write a Julia function that computes the first `n` terms of the Fibonacci sequence, returning these terms in an array.

```
function fibonacci(n)
    arr = [0, 1]

    return arr
end
```

Fill in the code for the function.